linear regression

CSCI 270

let's formalize and generalize

model space

$$y = \theta_S x_S + \theta_L x_L$$

loss function

$$L(\theta_S, \theta_L) = (.72\theta_S + .06\theta_L - 20)^2 + (.34\theta_S + .25\theta_L - 28)^2 + (.17\theta_S + .57\theta_L - 41)^2$$

training data

	x_S saturation	x_L lightness	y age
	.72	.06	20
	.34	.25	28
	.17	.57	41

optimization

$$\underset{\theta_S, \theta_L}{\operatorname{argmin}} L(\theta_S, \theta_L)$$

test data

let's formalize and generalize

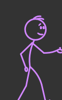
starting here

training data

	x_S saturation	x_L lightness	y age
	.72	.06	20
	.34	.25	28
	.17	.57	41

let's formalize and generalize

training data

	x_S saturation	x_L lightness	y age
	.72	.06	20
	.34	.25	28
	.17	.57	41

training instances

we will number each instance

training data

instance	x_S saturation	x_L lightness	y age
1	.72	.06	20
2	.34	.25	28
3	.17	.57	41

training data

instance	x_S saturation	x_L lightness	y age
1	.72	.06	20
2	.34	.25	28
3	.17	.57	41

features

we will number each feature

training data

instance	x_1 saturation	x_2 lightness	y age
1	.72	.06	20
2	.34	.25	28
3	.17	.57	41

features

we will almost always include a constant feature called the bias

training data

instance	x_1 bias	x_2 saturation	x_3 lightness	y age
1	1	.72	.06	20
2	1	.34	.25	28
3	1	.17	.57	41

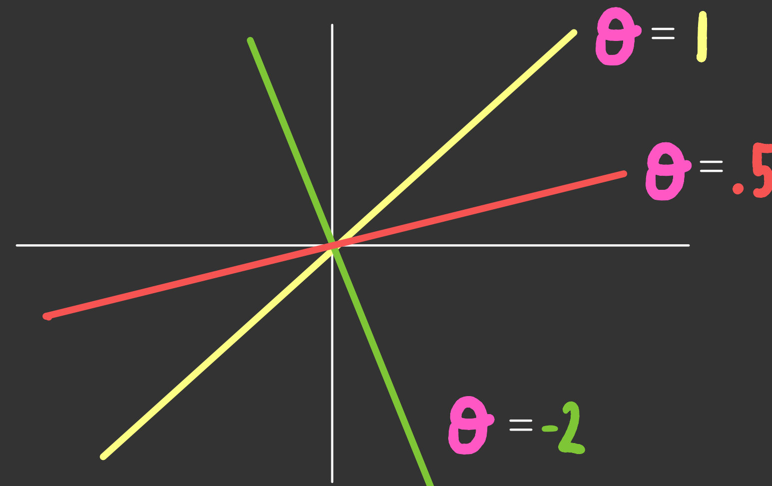
why?

if our model is

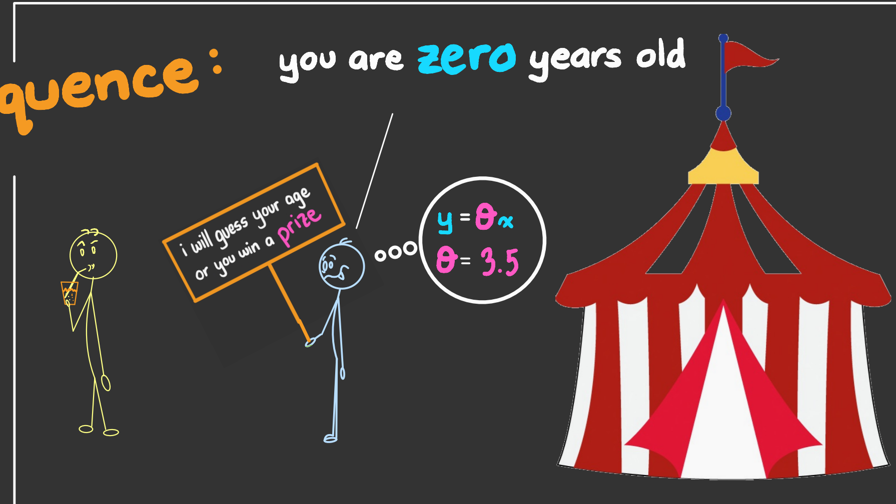
$$y = \theta x$$

then we are restricted to models that go through the **origin**

	x	y
	5	20
	8	28
	12	41



consequence:



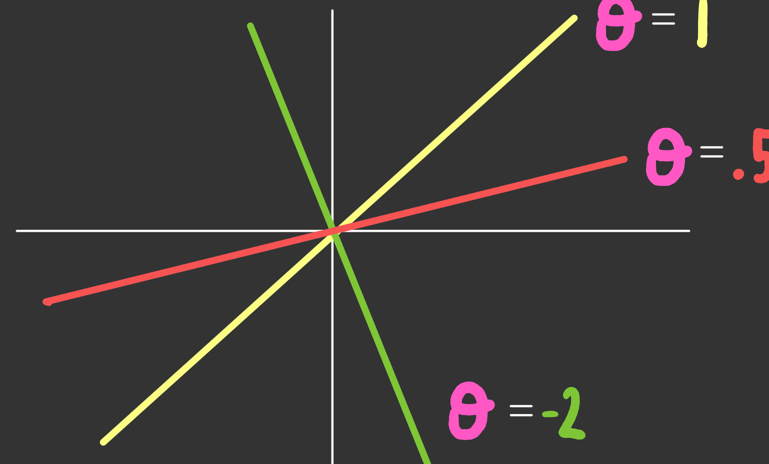
if our model is

$$y = \theta x$$

then we are restricted to models that go through the

origin

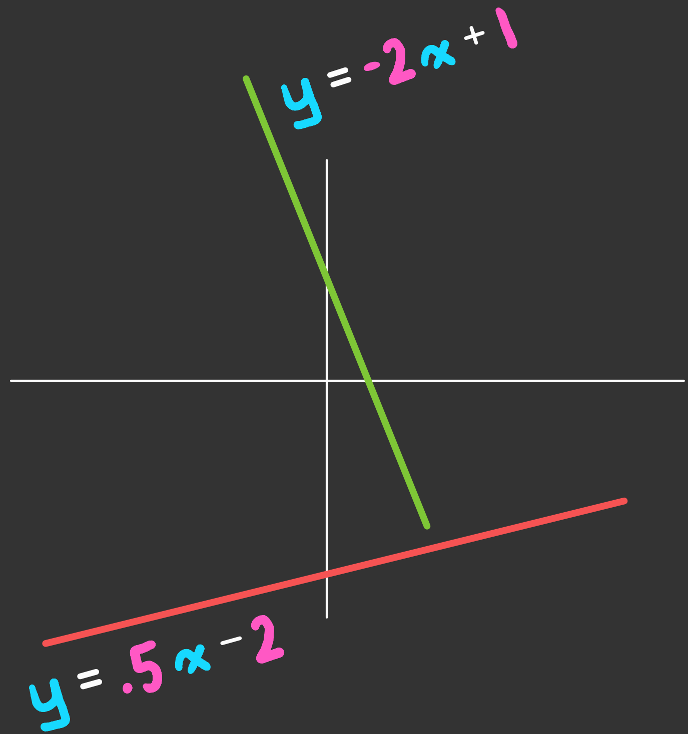
$y = ax$ a



patch:

include another parameter

$$y = \theta x + b$$



this is the same as including a feature whose value is always 1

$$y = \theta x + b$$

$$y = \theta_1 x_1 + \theta_2 x_2$$

always equals 1

we refer to the variable we want to predict as the outcome

training data

	features			outcome
instance	x_1 bias	x_2 saturation	x_3 lightness	y age
1	1	.72	.06	20
2	1	.34	.25	28
3	1	.17	.57	41

in general, there are N instances and D features

training data

instance	features				outcome
	x_1	x_2	$\cdots$	x_D	y
1	1	.72	$\cdots$	.06	20
2	1	.34	$\cdots$	.25	28
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
N	1	.17	$\cdots$	.57	41

in general, there are N instances and D features

training data

instance	features				outcome
	x_1	x_2	$\cdots$	x_D	y
1	$x_1^{(1)}$	$x_2^{(1)}$	$\cdots$	$x_D^{(1)}$	$y^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$\cdots$	$x_D^{(2)}$	$y^{(2)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
N	$x_1^{(N)}$	$x_2^{(N)}$	$\cdots$	$x_D^{(N)}$	$y^{(N)}$

let's formalize and generalize

next up

model space

$$y = \theta_S x_S + \theta_L x_L$$

loss function

$$L(\theta_S, \theta_L) = (.72\theta_S + .06\theta_L - 20)^2 + (.34\theta_S + .25\theta_L - 28)^2 + (.17\theta_S + .57\theta_L - 41)^2$$

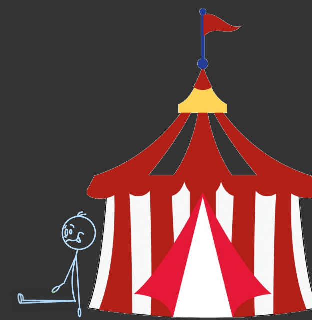
training data ✓

instance	x_1	x_2	$\cdots$	x_D	y
	features				outcome
1	$x_1^{(1)}$	$x_2^{(1)}$	$\cdots$	$x_D^{(1)}$	$y^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$\cdots$	$x_D^{(2)}$	$y^{(2)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
N	$x_1^{(N)}$	$x_2^{(N)}$	$\cdots$	$x_D^{(N)}$	$y^{(N)}$

optimization

$$\underset{\theta_S, \theta_L}{\operatorname{argmin}} L(\theta_S, \theta_L)$$

test data



training data

instance	features				outcome
	x_1	x_2	$\cdots$	x_D	y
1	$x_1^{(1)}$	$x_2^{(1)}$	$\cdots$	$x_D^{(1)}$	$y^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$\cdots$	$x_D^{(2)}$	$y^{(2)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
N	$x_1^{(N)}$	$x_2^{(N)}$	$\cdots$	$x_D^{(N)}$	$y^{(N)}$

let's generalize

model space

$$y = \theta_S x_S + \theta_L x_L$$

let's generalize

training data

instance	features				outcome
	x_1	x_2	$\cdots$	x_D	y
1	$x_1^{(1)}$	$x_2^{(1)}$	$\cdots$	$x_D^{(1)}$	$y^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$\cdots$	$x_D^{(2)}$	$y^{(2)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
N	$x_1^{(N)}$	$x_2^{(N)}$	$\cdots$	$x_D^{(N)}$	$y^{(N)}$

model space

$$y = \theta_1 x_1 + \cdots + \theta_D x_D$$

training data

instance	features				outcome
	x_1	x_2	$\cdots$	x_D	y
1	$x_1^{(1)}$	$x_2^{(1)}$	$\cdots$	$x_D^{(1)}$	$y^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$\cdots$	$x_D^{(2)}$	$y^{(2)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
N	$x_1^{(N)}$	$x_2^{(N)}$	$\cdots$	$x_D^{(N)}$	$y^{(N)}$

let's generalize

model space

$$y = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_D \end{bmatrix}$$

let's formalize and generalize

model space ✓

$$y = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_D \end{bmatrix}$$

training data ✓

instance	features x_1	x_2	$\cdots$	x_D	outcome y
1	$x_1^{(1)}$	$x_2^{(1)}$	$\cdots$	$x_D^{(1)}$	$y^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$\cdots$	$x_D^{(2)}$	$y^{(2)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
N	$x_1^{(N)}$	$x_2^{(N)}$	$\cdots$	$x_D^{(N)}$	$y^{(N)}$

loss function

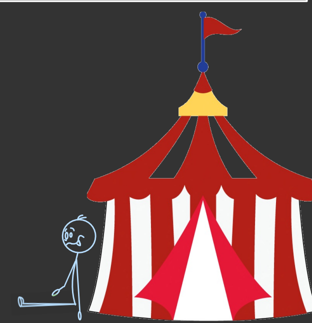
$$L(\theta_S, \theta_L) = (.72\theta_S + .06\theta_L - 20)^2 + (.34\theta_S + .25\theta_L - 28)^2 + (.17\theta_S + .57\theta_L - 41)^2$$

next up

optimization

$$\underset{\theta_S, \theta_L}{\operatorname{argmin}} L(\theta_S, \theta_L)$$

test data



recall: we want the loss function to quantify our embarrassment if we make predictions using specific **parameter** values

	features				outcome	prediction
instance	x_1	x_2	$\cdots$	x_D	y	$\hat{y}$
1	$x_1^{(1)}$	$x_2^{(1)}$	$\cdots$	$x_D^{(1)}$	$y^{(1)}$	$\theta \cdot x^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$\cdots$	$x_D^{(2)}$	$y^{(2)}$	$\theta \cdot x^{(2)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$	$\vdots$
N	$x_1^{(N)}$	$x_2^{(N)}$	$\cdots$	$x_D^{(N)}$	$y^{(N)}$	$\theta \cdot x^{(N)}$

$$\begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix}$$

parameter vector

we want our predictions to be close to the observed outcomes

instance	features				outcome	prediction
	x_1	x_2	$\cdots$	x_D	y	$\hat{y}$
1	$x_1^{(1)}$	$x_2^{(1)}$	$\cdots$	$x_D^{(1)}$	$y^{(1)}$	$\approx \theta \cdot x^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$\cdots$	$x_D^{(2)}$	$y^{(2)}$	$\approx \theta \cdot x^{(2)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$	$\vdots$
N	$x_1^{(N)}$	$x_2^{(N)}$	$\cdots$	$x_D^{(N)}$	$y^{(N)}$	$\approx \theta \cdot x^{(N)}$

$$\begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix}$$

parameter vector

$$\text{e.g.}\quad L(\theta) = \sum_{n=1}^{N} \left(\theta \cdot x^{(n)} - y^{(n)}\right)^2$$

	features				outcome	prediction
instance	x_1	x_2	$\cdots$	x_D	y	$\hat{y}$
1	$x_1^{(1)}$	$x_2^{(1)}$	$\cdots$	$x_D^{(1)}$	$y^{(1)}$	$\approx \theta \cdot x^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$\cdots$	$x_D^{(2)}$	$y^{(2)}$	$\approx \theta \cdot x^{(2)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$	$\vdots$
N	$x_1^{(N)}$	$x_2^{(N)}$	$\cdots$	$x_D^{(N)}$	$y^{(N)}$	$\approx \theta \cdot x^{(N)}$

$$\begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix}$$

parameter vector

$$\text{e.g. } L(\theta) = \sum_{n=1}^{N} \left(\theta \cdot x^{(n)} - y^{(n)}\right)^2$$

why squared?

- the loss is always non-negative
- the loss is zero if the predictions perfectly match the observed outcomes
- the loss is easy to differentiate

linear regression

model space ✓

$$y = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_D \end{bmatrix}$$

loss function ✓

$$L(\theta) = \sum_{n=1}^{N} \left(\theta \cdot x^{(n)} - y^{(n)}\right)^2$$

optimization ✓

$$\underset{\theta}{\operatorname{argmin}}\; L(\theta)$$

training data ✓

instance	features				outcome
	x_1	x_2	$\cdots$	x_D	y
1	$x_1^{(1)}$	$x_2^{(1)}$	$\cdots$	$x_D^{(1)}$	$y^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$\cdots$	$x_D^{(2)}$	$y^{(2)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
N	$x_1^{(N)}$	$x_2^{(N)}$	$\cdots$	$x_D^{(N)}$	$y^{(N)}$

test data